TECHLIFT – 100-Question Technical Quiz Bank

Target: 1st-year students
Recommended use: Select 40 questions from this 100-question bank for the actual Stage 1 quiz.
Difficulty: Beginner → Moderate
Suggested duration for 40 questions: 30 minutes
Format: MCQ, 4 options, 1 correct answer.

The topics are aligned with your TECHLIFT objective of testing **computer fundamentals, programming concepts, logical reasoning and technology awareness**.

SECTION A — COMPUTER FUNDAMENTALS

Questions 1–20

Q1. What is the main function of the CPU?

A. Store files permanently
B. Execute instructions and process data
C. Display images
D. Connect to the internet

Answer: B — Execute instructions and process data
Explanation: The CPU executes program instructions and performs calculations and logical operations.

Q2. Which of the following is volatile memory?

A. ROM
B. SSD
C. RAM
D. Hard Disk

Answer: C — RAM
Explanation: RAM temporarily stores data and instructions while a computer is running. Its contents are lost when power is removed.

Q3. Which device is primarily used to enter text into a computer?

A. Monitor
B. Keyboard
C. Speaker
D. Projector

Answer: B — Keyboard
Explanation: A keyboard is an input device used to enter characters, numbers and commands.

Q4. Which of the following is an output device?

A. Mouse
B. Keyboard
C. Scanner
D. Monitor

Answer: D — Monitor
Explanation: A monitor displays visual information generated by the computer.

Q5. What does OS stand for?

A. Online System
B. Operating System
C. Output Service
D. Open Software

Answer: B — Operating System
Explanation: An operating system manages computer hardware and provides services for applications.

Q6. Which of the following is an operating system?

A. Python
B. Windows
C. Google Chrome
D. MySQL

Answer: B — Windows
Explanation: Windows is an operating system. Python is a programming language, Chrome is a browser, and MySQL is a database management system.

Q7. Which unit is commonly used to represent computer storage?

A. Liter
B. Byte
C. Volt
D. Hertz

Answer: B — Byte
Explanation: Digital storage capacity is commonly measured in bytes and larger units such as KB, MB, GB and TB.

Q8. How many bits are there in one byte?

A. 2
B. 4
C. 8
D. 16

Answer: C — 8
Explanation: One byte consists of 8 binary bits.

Q9. What is the binary representation of decimal 5?

A. 101
B. 110
C. 111
D. 100

Answer: A — 101
Explanation: 5 = 4 + 1, so its binary representation is 101.

Q10. Which component performs arithmetic and logical operations inside the CPU?

A. ALU
B. RAM
C. SSD
D. NIC

Answer: A — ALU
Explanation: The Arithmetic Logic Unit performs arithmetic calculations and logical comparisons.

Q11. What does RAM stand for?

A. Read Access Memory
B. Random Access Memory
C. Rapid Application Memory
D. Remote Access Module

Answer: B — Random Access Memory
Explanation: RAM is the computer's primary temporary working memory.

Q12. Which storage device generally has no moving mechanical parts?

A. HDD
B. SSD
C. Floppy Disk
D. Magnetic Tape

Answer: B — SSD
Explanation: Solid-State Drives use flash memory and have no mechanical spinning disks.

Q13. Which component is mainly responsible for rendering graphics?

A. GPU
B. PSU
C. Keyboard
D. BIOS

Answer: A — GPU
Explanation: A Graphics Processing Unit is designed to process and render graphical data efficiently.

Q14. What is the purpose of a file extension such as .pdf?

A. It identifies the file type
B. It increases file size
C. It encrypts the file
D. It deletes the file

Answer: A — It identifies the file type
Explanation: File extensions commonly indicate the format or type of a file.

Q15. Which shortcut is commonly used to copy selected content in Windows?

A. Ctrl + X
B. Ctrl + V
C. Ctrl + C
D. Ctrl + Z

Answer: C — Ctrl + C
Explanation: Ctrl+C copies selected content to the clipboard.

Q16. Which shortcut is commonly used to paste copied content?

A. Ctrl + P
B. Ctrl + V
C. Ctrl + S
D. Ctrl + A

Answer: B — Ctrl + V
Explanation: Ctrl+V pastes content from the clipboard.

Q17. Which shortcut generally saves a document?

A. Ctrl + S
B. Ctrl + D
C. Ctrl + E
D. Ctrl + R

Answer: A — Ctrl + S
Explanation: Ctrl+S is the common save shortcut in many applications.

Q18. What does GUI stand for?

A. General User Input
B. Graphical User Interface
C. Graphic Utility Integration
D. General Utility Interface

Answer: B — Graphical User Interface
Explanation: A GUI allows users to interact with software using visual elements such as windows, icons and menus.

Q19. Which number system does a computer fundamentally use?

A. Decimal
B. Binary
C. Octal only
D. Roman

Answer: B — Binary
Explanation: Digital computers represent information internally using binary values, commonly 0 and 1.

Q20. Which of the following is system software?

A. Operating system
B. Presentation
C. Spreadsheet
D. Photo

Answer: A — Operating system
Explanation: System software manages hardware and provides a platform for application software.

SECTION B — PYTHON & PROGRAMMING BASICS

Questions 21–45

Q21. Which of the following is a programming language?

A. Python
B. Windows
C. Chrome
D. Linux

Answer: A — Python
Explanation: Python is a high-level general-purpose programming language.

Q22. Which symbol is used for a single-line comment in Python?

A. //
B. #
C. <!-- -->
D. /* */

Answer: B — #
Explanation: In Python, text following # on a line is treated as a comment.

Q23. Which function is used to display output in Python?

A. display()
B. output()
C. print()
D. show()

Answer: C — print()
Explanation: Python's print() function displays values or text to standard output.

Q24. Which function is commonly used to receive user input in Python?

A. scan()
B. input()
C. read()
D. get()

Answer: B — input()
Explanation: input() reads a line of user input.

Q25. Which of the following is an integer in Python?

A. 10
B. 10.5
C. "10"
D. True

Answer: A — 10
Explanation: 10 is an integer. 10.5 is a floating-point value and "10" is a string.

Q26. Which data type represents text in Python?

A. int
B. float
C. str
D. bool

Answer: C — str
Explanation: Python uses the str type for sequences of text characters.

Q27. What is the output of print(2 + 3)?

A. 23
B. 5
C. 6
D. Error

Answer: B — 5
Explanation: The + operator performs numerical addition when both operands are numbers.

Q28. What is the output of print(10 // 3)?

A. 3
B. 3.33
C. 1
D. 4

Answer: A — 3
Explanation: // performs floor division. 10 divided by 3 gives a floor result of 3.

Q29. Which operator is used for exponentiation in Python?

A. ^
B. **
C. //
D. %%

**Answer: B — ** **
Explanation: Python uses ** for exponentiation, such as 2 ** 3, which gives 8.

Q30. Which of the following is a valid Python variable name?

A. 2name
B. student-name
C. student_name
D. class

Answer: C — student_name
Explanation: Python identifiers cannot begin with a digit or contain a hyphen. class is a reserved keyword.

Q31. Which keyword is used for a condition in Python?

A. if
B. check
C. condition
D. when

Answer: A — if
Explanation: The if statement executes a block when its condition is true.

Q32. Which keyword is used when an if condition is false and another condition needs to be checked?

A. otherwise
B. elif
C. elseif
D. next

Answer: B — elif
Explanation: elif allows another condition to be checked after a preceding if or elif condition is false.

Q33. Which loop is commonly used when the number of iterations is determined by a sequence or range?

A. for
B. repeat
C. loop
D. iterate

Answer: A — for
Explanation: Python's for loop iterates over items in an iterable such as a list or range.

Q34. Which loop continues while a specified condition remains true?

A. for
B. while
C. repeat
D. until

Answer: B — while
Explanation: A while loop repeatedly executes its body while its condition evaluates to true.

Q35. What is the output of print(len("Python"))?

A. 5
B. 6
C. 7
D. Error

Answer: B — 6
Explanation: "Python" contains six characters.

Q36. Which data structure stores an ordered collection that can be modified?

A. Tuple
B. List
C. String
D. Integer

Answer: B — List
Explanation: Python lists are ordered, mutable collections.

Q37. Which brackets are used to create a Python list?

A. ()
B. {}
C. []
D. <>

Answer: C — []
Explanation: Lists are represented using square brackets, for example [1, 2, 3].

Q38. Which brackets are commonly used to create a tuple?

A. []
B. ()
C. {}
D. <>

Answer: B — ()
Explanation: Tuples are commonly written using parentheses, such as (1, 2, 3).

Q39. Which Python data structure stores key-value pairs?

A. List
B. Tuple
C. Dictionary
D. Set

Answer: C — Dictionary
Explanation: A dictionary maps keys to values.

Q40. What is the output of print(10 > 5)?

A. 10
B. 5
C. True
D. False

Answer: C — True
Explanation: 10 is greater than 5, so the comparison evaluates to True.

Q41. Which keyword is used to define a function in Python?

A. function
B. define
C. def
D. func

Answer: C — def
Explanation: Python uses the def keyword to define a function.

Q42. What is the purpose of a function?

A. To permanently store files
B. To group reusable instructions
C. To increase monitor resolution
D. To connect to Wi-Fi

Answer: B — To group reusable instructions
Explanation: Functions organize code into reusable blocks that can be called when needed.

Q43. What does the break statement do inside a loop?

A. Skips one iteration and continues
B. Terminates the loop
C. Restarts the loop
D. Pauses the computer

Answer: B — Terminates the loop
Explanation: break immediately exits the nearest enclosing loop.

Q44. What does the continue statement do inside a loop?

A. Terminates the program
B. Exits the loop completely
C. Skips the current iteration and continues
D. Repeats the previous iteration

Answer: C — Skips the current iteration and continues
Explanation: continue moves execution to the next iteration of the loop.

Q45. What is the output of print("Data" + "Science")?

A. Data Science
B. DataScience
C. Data+Science
D. Error

Answer: B — DataScience
Explanation: The + operator concatenates the two strings without automatically inserting a space.

SECTION C — LOGICAL REASONING & OUTPUT PREDICTION

Questions 46–60

Q46. What is the next number in the sequence: 2, 4, 8, 16, ___?

A. 20
B. 24
C. 32
D. 36

Answer: C — 32
Explanation: Each number is multiplied by 2.

Q47. If 5 machines produce 5 items in 5 minutes, how many minutes would 1 machine take to produce 1 item at the same rate?

A. 1 minute
B. 5 minutes
C. 10 minutes
D. 25 minutes

Answer: B — 5 minutes
Explanation: Each machine produces one item in 5 minutes under the given rate.

Q48. Which number does not belong to the group?

A. 2
B. 4
C. 8
D. 15

Answer: D — 15
Explanation: 2, 4 and 8 are powers of 2; 15 is not.

Q49. If A = 1, B = 2, C = 3, what is the value of CAT?

A. 21
B. 22
C. 24
D. 26

Answer: C — 24
Explanation: C=3, A=1, T=20. Therefore, 3 + 1 + 20 = 24.

Q50. What is the next number: 1, 4, 9, 16, 25, ___?

A. 30
B. 36
C. 40
D. 49

Answer: B — 36
Explanation: These are squares: 1^2, 2^2, 3^2, 4^2, 5^2; the next is $6^2 = 36$.

Q51. If a program executes instructions in sequence, what generally happens first?

A. The first instruction is executed
B. The last instruction is executed
C. All instructions execute simultaneously
D. The program automatically skips instructions

Answer: A — The first instruction is executed
Explanation: Sequential execution normally follows the order of instructions unless control-flow statements change it.

Q52. What is the output of this Python expression?

3 * 4 + 2

A. 18
B. 14
C. 20
D. 24

Answer: B — 14
Explanation: Multiplication has higher precedence than addition: 3 × 4 + 2 = 14.

Q53. What is the output of 10 - 2 * 3?

A. 24
B. 4
C. 8
D. 6

Answer: B — 4
Explanation: Multiplication is performed first: 10 − (2 × 3) = 4.

Q54. What is the output of 2 ** 3 in Python?

A. 6
B. 8
C. 9
D. 5

Answer: B — 8
Explanation: ** is the exponentiation operator, so $2^3 = 8$.

Q55. If a number is divisible by both 2 and 3, which number must divide it?

A. 4
B. 5
C. 6
D. 9

Answer: C — 6
Explanation: A number divisible by both 2 and 3 is divisible by their least common multiple, 6.

Q56. A program has 10 iterations and skips every second iteration starting with iteration 2. How many iterations execute their main body?

A. 4
B. 5
C. 6
D. 10

Answer: B — 5
Explanation: Iterations 1, 3, 5, 7 and 9 execute their main body.

Q57. Which value is the largest?

A. 0.5
B. 0.05
C. 0.505
D. 0.055

Answer: C — 0.505
Explanation: Comparing the decimal values, 0.505 is greater than the other three.

Q58. If today is Monday, what day will it be after 10 days?

A. Wednesday
B. Thursday
C. Friday
D. Saturday

Answer: B — Thursday
Explanation: Seven days later is Monday; three additional days gives Thursday.

Q59. A binary number contains only which two digits?

A. 0 and 1
B. 1 and 2
C. 0 and 9
D. 2 and 8

Answer: A — 0 and 1
Explanation: Binary is a base-2 number system using only 0 and 1.

Q60. Which logical operator returns True when at least one of its conditions is True?

A. AND
B. OR
C. NOT
D. XOR only

Answer: B — OR
Explanation: In standard Boolean logic, OR evaluates to true when one or more inputs are true.

SECTION D — DATABASE & SQL BASICS

Questions 61–70

Q61. What does DBMS stand for?

A. Data Backup Management System
B. Database Management System
C. Digital Base Management Service
D. Database Machine Software

Answer: B — Database Management System
Explanation: A DBMS is software used to create, manage and interact with databases.

Q62. Which of the following is a relational database system?

A. MySQL
B. Photoshop
C. Python
D. Windows

Answer: A — MySQL
Explanation: MySQL is a relational database management system.

Q63. Which SQL command is used to retrieve data?

A. GET
B. SELECT
C. FETCHALL
D. SHOWDATA

Answer: B — SELECT
Explanation: SELECT is used to retrieve data from database tables.

Q64. Which SQL command is used to add a new row to a table?

A. INSERT
B. ADDROW
C. APPEND
D. CREATE

Answer: A — INSERT
Explanation: The SQL INSERT statement adds new records to a table.

Q65. Which SQL command is used to modify existing records?

A. CHANGE
B. MODIFY
C. UPDATE
D. ALTERDATA

Answer: C — UPDATE
Explanation: UPDATE modifies values in existing rows.

Q66. Which SQL command removes rows from a table?

A. REMOVE
B. DELETE
C. DROPROW
D. CLEAR

Answer: B — DELETE
Explanation: DELETE removes selected rows from a table.

Q67. What is a primary key?

A. A column that uniquely identifies rows
B. A password for a database
C. A backup copy of a database
D. A column containing only text

Answer: A — A column that uniquely identifies rows
Explanation: A primary key uniquely identifies each record in a relational table.

Q68. Which SQL clause is used to filter rows?

A. SORT
B. WHERE
C. FILTERBY
D. CONDITION

Answer: B — WHERE
Explanation: WHERE specifies conditions that rows must satisfy.

Q69. Which SQL clause is commonly used to sort query results?

A. ORDER BY
B. SORT BY
C. ARRANGE
D. GROUP SORT

Answer: A — ORDER BY
Explanation: ORDER BY sorts query results in ascending or descending order.

Q70. In a relational database, what is a table?

A. A collection of related rows and columns
B. A computer monitor
C. A programming loop
D. A network connection

Answer: A — A collection of related rows and columns
Explanation: Relational databases organize data into tables consisting of rows and columns.

SECTION E — INTERNET, NETWORKING & CYBERSECURITY

Questions 71–80

Q71. What does URL stand for?

A. Universal Resource Link
B. Uniform Resource Locator
C. Unified Routing Language
D. User Resource Location

Answer: B — Uniform Resource Locator
Explanation: A URL identifies the location of a resource on the internet.

Q72. Which protocol is commonly used to access websites securely?

A. HTTP
B. HTTPS
C. FTP
D. SMTP

Answer: B — HTTPS
Explanation: HTTPS uses encryption through TLS to protect communication between a browser and a web server.

Q73. What is the main purpose of a web browser?

A. Compile Python code
B. Access and display web content
C. Store electricity
D. Manage computer hardware

Answer: B — Access and display web content
Explanation: Browsers retrieve and render web pages and other web resources.

Q74. Which of the following is a web browser?

A. Google Chrome
B. MySQL
C. Ubuntu
D. Python

Answer: A — Google Chrome
Explanation: Chrome is a web browser used to access websites and web applications.

Q75. What is an IP address used for?

A. Identifying a device/interface on a network
B. Increasing RAM
C. Encrypting every file
D. Formatting a hard disk

Answer: A — Identifying a device/interface on a network
Explanation: IP addresses are used for addressing and routing network traffic.

Q76. Which device forwards data between different networks?

A. Keyboard
B. Router
C. Monitor
D. Printer

Answer: B — Router
Explanation: Routers forward packets between networks based on network addressing.

Q77. What is phishing?

A. A method of improving Wi-Fi speed
B. A fraudulent attempt to obtain sensitive information
C. A type of computer processor
D. A data compression method

Answer: B — A fraudulent attempt to obtain sensitive information
Explanation: Phishing commonly uses deceptive messages or websites to trick users into revealing credentials or other sensitive information.

Q78. Which is the strongest password?

A. 12345678
B. password
C. Eshwar123
D. T9#qL7!vP2@x

Answer: D — T9#qL7!vP2@x
Explanation: A strong password should be difficult to guess and preferably use a combination of character types.

Q79. What does malware refer to?

A. Malicious software
B. Manual hardware
C. Mathematical software
D. Mail server

Answer: A — Malicious software
Explanation: Malware is software designed to harm systems, disrupt operations or perform unauthorized actions.

Q80. What is two-factor authentication?

A. Using two computers simultaneously
B. Using two independent verification factors
C. Having two passwords of the same type
D. Logging into two websites

Answer: B — Using two independent verification factors
Explanation: 2FA requires an additional verification factor beyond the primary credential, such as a password plus a code or security key.

SECTION F — DATA SCIENCE & AI AWARENESS

Questions 81–90

Q81. What is data science primarily concerned with?

A. Extracting useful knowledge and insights from data
B. Designing computer keyboards
C. Repairing monitors
D. Installing operating systems only

Answer: A — Extracting useful knowledge and insights from data
Explanation: Data science combines statistics, programming and analytical methods to extract useful insights from data.

Q82. Which Python library is widely used for numerical array operations?

A. NumPy
B. Flask
C. Django
D. Tkinter

Answer: A — NumPy
Explanation: NumPy provides efficient multidimensional arrays and numerical computing functionality.

Q83. Which Python library is widely used for tabular data analysis?

A. Pandas
B. Turtle
C. Pygame
D. Requests

Answer: A — Pandas
Explanation: Pandas provides structures such as DataFrame for working with structured and tabular data.

Q84. What is a dataset?

A. A collection of related data
B. A computer processor
C. A network cable
D. A programming language

Answer: A — A collection of related data
Explanation: A dataset is a structured or unstructured collection of data used for analysis, processing or modeling.

Q85. What is machine learning?

A. A method where systems learn patterns from data
B. A type of computer monitor
C. A database storage device
D. A programming keyboard

Answer: A — A method where systems learn patterns from data
Explanation: Machine-learning methods allow models to learn patterns or relationships from data for tasks such as prediction or classification.

Q86. Which is an example of supervised learning?

A. Classification using labeled training data
B. Randomly deleting data
C. Formatting a hard disk
D. Connecting two monitors

Answer: A — Classification using labeled training data
Explanation: Supervised learning uses examples with known target labels to learn a mapping from inputs to outputs.

Q87. What is the purpose of training data in machine learning?

A. To help a model learn patterns
B. To physically train a computer user
C. To increase monitor brightness
D. To install an operating system

Answer: A — To help a model learn patterns
Explanation: Training data provides examples from which a machine-learning model learns relationships.

Q88. What does AI stand for?

A. Automated Internet
B. Artificial Intelligence
C. Advanced Interface
D. Applied Information

Answer: B — Artificial Intelligence
Explanation: Artificial Intelligence refers broadly to computational systems designed to perform tasks associated with aspects of human intelligence.

Q89. Which of the following is an example of generative AI?

A. A system that generates a new text response from a prompt
B. A calculator performing addition
C. A keyboard entering characters
D. A hard disk storing files

Answer: A — A system that generates a new text response from a prompt
Explanation: Generative AI systems can produce new content such as text, images, audio or code based on learned patterns and input instructions.

Q90. What is data visualization?

A. Representing data graphically
B. Encrypting a database
C. Deleting duplicate files
D. Installing software

Answer: A — Representing data graphically
Explanation: Data visualization uses charts, graphs and other visual forms to communicate patterns and relationships in data.

SECTION G — GENERAL TECHNOLOGY & COMPUTING AWARENESS

Questions 91–100

Q91. What does HTML primarily define?

A. Structure of web pages
B. Database passwords
C. CPU instructions
D. Antivirus signatures

Answer: A — Structure of web pages
Explanation: HTML provides the structural markup used to organize content on web pages.

Q92. What does CSS primarily control?

A. Web page presentation and styling
B. Database transactions
C. CPU temperature
D. File compression

Answer: A — Web page presentation and styling
Explanation: CSS controls visual presentation such as layout, fonts, spacing and other styles.

Q93. What does API stand for?

A. Application Programming Interface
B. Advanced Program Internet
C. Application Process Integration
D. Automated Programming Input

Answer: A — Application Programming Interface
Explanation: An API defines a way for software components to communicate and interact.

Q94. What is cloud computing?

A. Delivering computing resources over a network, typically the internet
B. Storing files only on USB drives
C. Repairing cloud servers manually
D. Using computers only during rainy weather

Answer: A — Delivering computing resources over a network, typically the internet
Explanation: Cloud computing provides resources such as computing power, storage and services over a network.

Q95. What does Git primarily help developers manage?

A. Versions of source code and related files
B. Computer temperature
C. Internet bandwidth
D. Monitor brightness

Answer: A — Versions of source code and related files
Explanation: Git is a distributed version-control system used to track changes and collaborate on code.

Q96. What is an algorithm?

A. A step-by-step procedure for solving a problem
B. A physical computer component
C. A type of monitor
D. A network cable

Answer: A — A step-by-step procedure for solving a problem
Explanation: An algorithm defines a sequence of steps for accomplishing a task or solving a problem.

Q97. Which programming concept allows the same operation to behave differently for different objects or inputs?

A. Polymorphism
B. Compilation

C. Encryption
D. Formatting

Answer: A — Polymorphism
Explanation: Polymorphism allows a common interface or operation to have different implementations or behaviors.

Q98. What is debugging?

A. Finding and fixing errors in a program
B. Increasing internet speed
C. Designing a logo
D. Installing a monitor

Answer: A — Finding and fixing errors in a program
Explanation: Debugging involves identifying the cause of program errors and correcting them.

Q99. What is open-source software?

A. Software whose source code is made available under terms that permit specified use, modification and redistribution
B. Software that never requires electricity
C. Software that only works offline
D. Software that cannot be modified

Answer: A — Software whose source code is made available under terms that permit specified use, modification and redistribution
Explanation: Open-source licenses provide permissions concerning use, inspection, modification and/or redistribution, subject to the specific license terms.

Q100. Which skill is most important when solving an unfamiliar programming problem?

A. Memorizing every programming language
B. Breaking the problem into smaller logical steps
C. Typing as fast as possible
D. Using the longest possible code

Answer: B — Breaking the problem into smaller logical steps
Explanation: Decomposition helps convert a complex problem into smaller, manageable tasks that can be solved systematically.

🔑 **MASTER ANSWER KEY**

For your organizing team, here is the quick answer key:

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	26	C	51	A	76	B
2	C	27	B	52	B	77	B
3	B	28	A	53	B	78	D
4	D	29	B	54	B	79	A
5	B	30	C	55	C	80	B
6	B	31	A	56	B	81	A
7	B	32	B	57	C	82	A
8	C	33	A	58	B	83	A
9	A	34	B	59	A	84	A
10	A	35	B	60	B	85	A
11	B	36	B	61	B	86	A
12	B	37	C	62	A	87	A
13	A	38	B	63	B	88	B
14	A	39	C	64	A	89	A
15	C	40	C	65	C	90	A
16	B	41	C	66	B	91	A
17	A	42	B	67	A	92	A
18	B	43	B	68	B	93	A
19	B	44	C	69	A	94	A
20	A	45	B	70	A	95	A
21	A	46	C	71	B	96	A
22	B	47	B	72	B	97	A
23	C	48	D	73	B	98	A
24	B	49	C	74	A	99	A
25	A	50	B	75	A	100	B